SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	ASEA
Project / Solution Name	Path.AI: find your path in the AI era
Name (Leader)	Almat Sakiidulla
Email & Phone (Leader)	sat200707@mail.ru, +77072203626 (WhatsApp)
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://pathai.lol
Code Repository URL (public / access granted to judges)	https://github.com/Almatikooo/Competition_Repo (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Browser → FastAPI backend → deterministic engine and data → AI model → accounts → hosting:

1. Interface. A no-build JavaScript single-page app in English, Malay and Chinese: wizard, matches, plan, Learn, Opportunities, AI Chat, My path, achievements and Rewards.

2. Backend. Python with FastAPI and Pydantic. A deterministic engine computes skill fit, demand under three AI-adoption scenarios and the learning plan.

3. AI. Qwen3.8-27B-FP8 on vLLM on a RunPod RTX A6000 (OpenAI-compatible API), started on demand. Groq hosts the same model as backup; 153 saved answers and rule-based text cover outages. Guards reject unquoted CV skills and any number not in the computed facts.

4. Data. Versioned JSON catalogs and dated caches in Git: 63 roles, 53 skills, 53 courses, 19,481 job postings, Hacker News and OpenAlex signals. SQLite holds the optional accounts.

5. Identity. Google, GitHub and Apple sign-in (OAuth 2.0 / OpenID Connect); signed, revocable session cookies.

6. Hosting. DigitalOcean VPS in Singapore behind Caddy (HTTPS, Let's Encrypt) under systemd; the same repository runs on Replit.

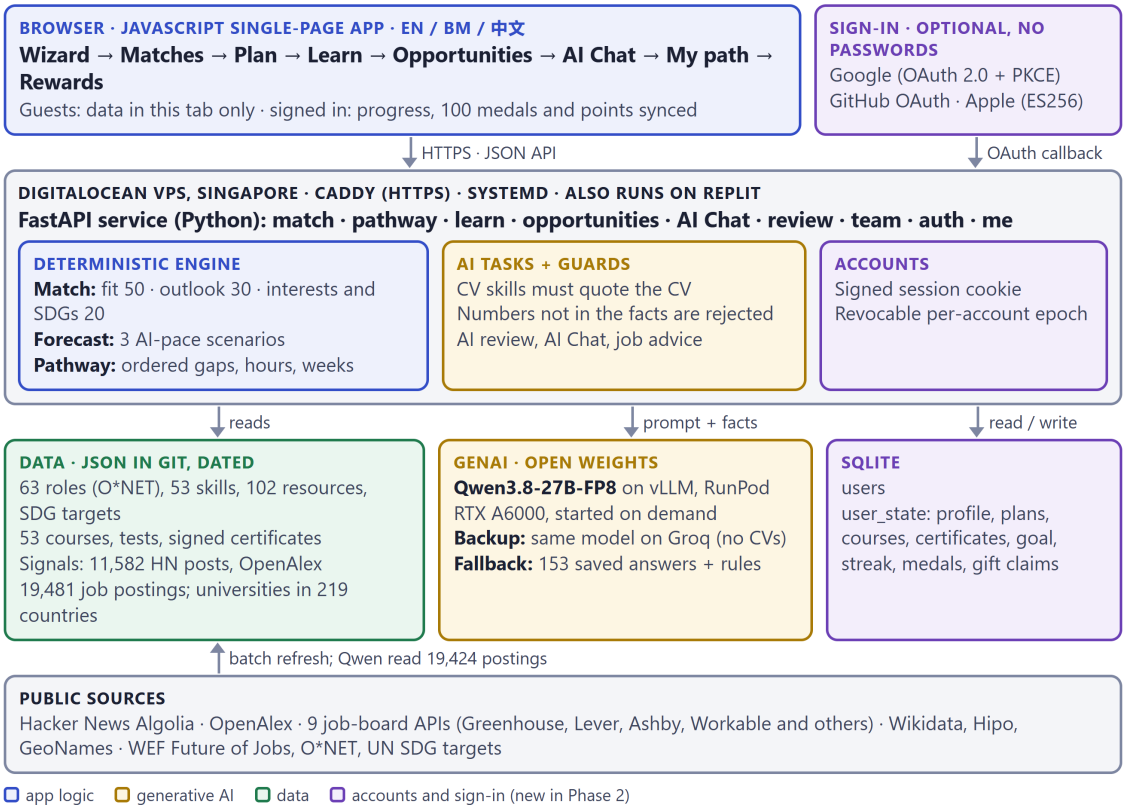


Figure 1. System block diagram (live at https://pathai.lol, commit 7907110, 30 October 2026).

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12; JavaScript (ES modules, no framework)	Backend language; frontend logic	PSF License; —
FastAPI ≥ 0.115	REST API framework	MIT
Uvicorn ≥ 0.30	ASGI web server	BSD-3-Clause
Pydantic ≥ 2.7	Input validation and schemas	MIT
httpx ≥ 0.27	Calls to data APIs, the model endpoint and OAuth providers	BSD-3-Clause

Name & Version	Purpose in Project	License
python-multipart ≥ 0.0.9; pypdf ≥ 4.2	CV upload; text from PDF CVs	Apache-2.0; BSD-3-Clause
cryptography ≥ 43	ES256 client secret for Sign in with Apple	Apache-2.0 or BSD-3-Clause
SQLite (Python sqlite3)	Accounts and synced progress	Public domain
pytest ≥ 8.0	162 automated tests	MIT
vLLM v0.31.0 (vllm/vllm-openai image)	Serves Qwen3.8-27B-FP8 on the RunPod GPU	Apache-2.0
Caddy 2	HTTPS reverse proxy on the server	Apache-2.0
Roboto (Google Fonts); flag-icons	Typography; country flags	SIL OFL 1.1; MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Qwen3.8-27B-FP8 via our vLLM endpoint on RunPod	CV skill extraction, job-posting facts, explanations, job advice, AI review and AI Chat	Sponsor-provided (RunPod credit)
Groq API (qwen/qwen3.8-27b)	Backup for AI answers while our GPU is off; never receives a CV	Own API key (not sponsor-provided)
RunPod REST API	Starting and stopping the GPU pod	Sponsor-provided (RunPod account)
Hacker News Search API (Algolia)	Monthly “Who is hiring?” posts: skill-demand signal and job postings	Free, public
OpenAlex API	Research share of AI capabilities (AI-pressure signal)	Free, public (CC0 data)
Greenhouse, Lever, Ashby, Workable job-board APIs	Open postings from company job boards	Free, public
Himalayas, Jobicy, Remotive, Remote OK, Arbeitnow APIs	Remote jobs and jobs in Germany	Free, public
Wikidata (QLever SPARQL), GeoNames	Universities by country; job locations to countries	Free, public
Google OAuth 2.0 / OpenID Connect; GitHub OAuth	Optional sign-in	Free
Sign in with Apple	Optional sign-in	Apple Developer Program membership
Google Fonts	Roboto typeface	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
DigitalOcean Droplet (Ubuntu 24.04, uvicorn under systemd)	Production hosting of https://pathai.lol: app, API and SQLite accounts database	1 GB RAM, Singapore; own paid plan
Caddy with Let's Encrypt	HTTPS certificates, reverse proxy, domain aliases	Free; on the same droplet
RunPod GPU Pod, Secure Cloud	vLLM + Qwen on 1× RTX A6000 (48 GB); 40 GB volume with the weights; on demand, a watchdog stops it after 20 idle minutes	$0.59/h, sponsor credit; region not specified
Groq Cloud	Hosted backup inference of the same open model	Own API key; region not specified
Replit	Team workspace, Replit Agent, one-click run (.replit); Phase 1 live prototype	Sponsor-provided
Hugging Face Hub	Download of the open model weights	Free
GitHub	Public repository and version history	Free
Google Cloud, GitHub and Apple developer consoles	OAuth clients for sign-in	Free; Apple Developer Program

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3.8-27B-FP8	Alibaba Qwen team; weights from Hugging Face	Pre-trained, not fine-tuned; self-hosted with vLLM on RunPod	Apache-2.0
qwen/qwen3.8-27b on Groq	Groq (the same open model, hosted)	Pre-trained; backup only	Apache-2.0 weights; Groq terms
Path.AI matching, forecast and pathway engine	Self-built (Team ASEA)	Deterministic formulas and rules; no training	Apache-2.0 (repository)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
HN “Who is hiring?” posts	hn.algolia.com; 36 monthly threads, 2023-11 to 2026-10 (11,582 posts)	Public (aggregate counts only)	Public API; Y Combinator terms	N
OpenAlex AI research	openalex.org; 5 AI capabilities, 2024H1–2026H1	Public	CC0	N
Job postings snapshot	213 company and remote job boards; 19,481 postings, facts only	Public	Each board's public API terms	N
Job facts	Read from 19,424 postings by Qwen (skills, degree, pay, …)	Self-generated (derived)	Apache-2.0 (repository)	N
Universities	Wikidata, Hipo university-domains-list, GeoNames; 219 countries	Public	CC0, MIT, CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET-SOC occupation codes	onetcenter.org; 63 roles	Public	CC BY 4.0	N
WEF Future of Jobs 2023/2025, ILO WP 140, Hays 2025	Official PDFs, cited at the page	Public (cited)	Cited with attribution; ILO CC BY 4.0	N
Open course content	OpenStax, Microsoft “for Beginners”, The Carpentries, BCcampus, GOV.UK, NIST and others; 53 courses, 755 lessons, 1,966 questions	Public	CC BY 4.0, CC BY-SA 4.0, MIT, OGL v3.0, CC0, US public domain	N
Gift photos	Wikimedia Commons; 30 photos, credited on the Rewards page	Public	CC0, CC BY, CC BY-SA, public domain	N
Demo personas and CVs	Self-created: 6 fictional people	Synthetic	Apache-2.0 (repository)	N (fictional)
Market survey	Google Form, 55 answers, 22–25 Oct 2026; market analysis only, not used by the app	Self-collected	Team's own	N (anonymous)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic)	Writing and refactoring code and tests, deployment scripts, README	Major
Claude (Anthropic)	Drafting submission documents, pitch deck and survey analysis	Moderate
Replit Agent (Replit)	Research: 27 job boards, checking 32 cited facts against WEF/ILO PDFs; course converter prototype	Moderate
ChatGPT / GitHub Copilot	Code completion and debugging (as declared in Phase 1)	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-assisted change was read and run by a team member before it was committed; commits Claude co-wrote carry a “Co-Authored-By” line. 162 automated tests run on each change, and the live site is checked by scripts (6 benchmark plans, 114 links). Facts found by Replit Agent were re-checked against the original PDFs and job boards before use, and numbers in our documents were recomputed from the code and data. The team takes full responsibility for all code, data and text.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core workspace and Replit Agent	Research agent (27 job boards found; 32 cited facts checked), team workspace with one-click run, Phase 1 live prototype
RunPod	$25 GPU credit (Secure Cloud RTX A6000)	Self-hosted Qwen3.8-27B-FP8 on vLLM: read 19,424 job postings (about 3 GPU hours) and writes the guarded AI answers; about $4.50 spent

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Neither is essential to the core. Matching, plans, courses, tests, certificates, opportunities, accounts and rewards run on our own server without them. RunPod powers the AI features; without it the same model on Groq, 153 saved answers and rule-based text take over, and the job facts already extracted stay in the repository. Replit sped up research and development; the app is now hosted on DigitalOcean, and the repository still runs on Replit with one click.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Guests: no account; profile, course progress and AI Chat stay in the browser tab and are sent only with the request that needs them.

Signed in (optional, Google, GitHub or Apple): name, email, picture and progress (profile, plans, courses, certificates, AI Chat, activity, medals, reward claims) are kept in SQLite on our Singapore server until the person deletes the account, which removes everything. No passwords are stored, OAuth tokens are used once, and sessions are HMAC-signed HttpOnly cookies revoked on sign-out.

CVs are read in memory and sent only to our own model on RunPod, never to Groq. The CV text is not stored; the model's answer (matched skills with short quotes) stays in the server's answer cache. Groq receives only computed facts. Reward claims are a demo with no payment or address data. Keys live in server environment variables, outside the repository.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Hallucination. Numbers not in the computed facts, links to unknown pages and unquoted CV skills are rejected; scores come from deterministic code; rule-based text is the fallback.

2. Data bias. Hacker News posts are global and tech-heavy, and only 166 postings are on-site in Malaysia. Outlooks show uncertainty bands, and a test keeps age, gender, ethnicity and location out of rankings.

3. Provider dependency and rate limits. RunPod may have no free GPU and Groq limits requests. Saved answers, the Groq backup and the rule-based path keep every page working.

4. Assumptions. Skill sensitivities to AI are team estimates; forecasts are labelled as estimates.

5. CV quotes in the answer cache. Planned fix: stop caching CV requests.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Almat Sakiidulla

30 October 2026